

Exercise 1

Let X be an $\text{AR}(\infty)$ process defined by:

$$X_t = w_t - \sum_{k=1}^{\infty} \lambda^{2k} X_{t-k},$$

where $w \sim (0, \sigma^2)$ and $0 < \lambda < 1$.

Write this process under the form $\Phi(B)X = \Theta(B)w$, with Φ and Θ polynomials (i.e. as finite sums of powers of B).

Correction: We identify easily the integral series:

$$w_t = (1 - \lambda^2 B)^{-1} X_t$$

which leads to:

$$w_t - \lambda^2 w_{t-1} = X_t$$

□

Exercise 2

Solve the $\text{ARMA}(1, 1)$ equation:

$$2X_t = X_{t-1} + 2w_t + w_{t-1}.$$

Correction: We can rewrite the equation as:

$$2X_t - X_{t-1} = 2w_t + w_{t-1}$$

That is, by using $(2 - B)^{-1} = \frac{1}{2} \sum_{i=1}^{\infty} \frac{1}{2^i} B^i$, we get:

$$\begin{aligned} X_t &= \left(\frac{1}{2} \sum_{i=1}^{\infty} \frac{1}{2^i} B^i \right) (2 + B) w_t = \left(\sum_{i=1}^{\infty} \frac{1}{2^i} B^i + \frac{1}{2} \sum_{i=1}^{\infty} \frac{1}{2^i} B^{i+1} \right) w_t \\ &= \left(\sum_{i=0}^{\infty} \frac{1}{2^i} B^i + \sum_{i=1}^{\infty} \frac{1}{2^{i+1}} B^{i+1} \right) w_t = \left(1 + \sum_{i=1}^{\infty} \frac{1}{2^{i-1}} B^i \right) w_t. \end{aligned}$$

The operator $1 + \sum_{i=1}^{\infty} \frac{1}{2^{i-1}} B^i$ is well defined as the series $\sum_{i=1}^{\infty} \frac{1}{2^{i-1}} x^i$ is absolutely convergent for any $|x| \leq 1$.

□

Exercise 3

Consider a regular ARMA(p, q) process (Y_t) driven by a Gaussian white noise $\varepsilon_t \sim (0, \sigma^2)$ associated with polynomials $\Phi(B) = (1 - \phi_1 B - \dots - \phi_p B^p)$ and $\Theta(B) = 1 - \theta_1 B - \dots - \theta_q B^q$.

1. Describe the general Box-Jenkins approach to forecasting Y_{n+h} at time n , $h \geq 1$.
2. Defining the error of the Box-Jenkins forecast Y_{n+h}^n as

$$e_n(h) = Y_{n+h} - Y_{n+h}^n,$$

find an expression for $\text{var}(e_n(h))$ and build a α -confidence interval for Y_{n+2} given $Y_1, \dots, Y_n$.

3. Consider a stationary ARMA model with a constant that is:

$$\Phi(B)Y_t = c + \Theta(B)\varepsilon_t.$$

Show that if there is no autoregressive part (i.e. $p = 0$) then $E[Y_t] = c$. Deduce $E[Y_t]$ in the general case.

4. Assume the model is now a stationary ARMA(1, 0), compute γ_k the autocovariance at lag k and show it is equal to:

$$\gamma_k = \gamma_0 \phi^k.$$

5. Let $\bar{Y} = n^{-1}(Y_1 + \dots + Y_n)$. Assuming that (Y_t) is stationary, show that the rescaled variance of the average $n\text{var}(\bar{Y})$ is

$$n\text{var}(\bar{Y}_n) = \gamma_0 + 2 \sum_{k=1}^{n-1} (1 - k/n) \gamma_k,$$

where $\gamma_k = \text{cov}(Y_t, Y_{t+k})$. If in addition Y_t follows a stationary ARMA(1, 0) defined by

$$Y_t = \phi Y_{t-1} + \varepsilon_t, \quad |\phi| < 1$$

show, using an expression for the covariance of the AR(1) process, that

$$n\text{var}(\bar{Y}_n) \rightarrow_{n \rightarrow \infty} \gamma_0 \left(\frac{1 + \phi}{1 - \phi} \right),$$

and describe the effect of autocorrelation in the variance of the average of an AR(1) process.

Correction:

1. We know that, since the process is regular, using Wold's decomposition, for some ψ :

$$\begin{aligned} Y_{n+h} &= \sum_{i=0}^{\infty} \psi_i \varepsilon_{n+h-i} \\ &= \psi_0 \varepsilon_{n+h} + \psi_1 \varepsilon_{n+h-1} + \cdots + \psi_h \varepsilon_n + \dots \end{aligned}$$

In the Box-Jenkins approach we set all future ε to 0, to minimize the mean squared forecasting error.

2. With the previous notation:

$$e_n(h) = Y_{n+h} - Y_{n+h}^n = \varepsilon_{n+h} + \psi_1 \varepsilon_{n+h-1} + \cdots + \psi_{h-1} \varepsilon_{n+1}.$$

thus by white-noisiness

$$\text{var}(e_n(h)) = \sigma^2(1 + \psi_1 + \cdots + \psi_{h-1}).$$

We can then build a confidence interval using the previous result for $h = 2$:

$$[Y_{n+2} \pm q_{\alpha/2} \sigma \sqrt{1 + \psi_1^2}],$$

where $q_{\alpha/2}$ is a quantile of rank $\alpha/2$ of the normal distribution.

3. If $\Phi(B) = 1$, we are in the situation where Wold's decomposition is trivial:

$$Y_t = c + \Theta(B)\varepsilon_t$$

so that $E[Y_t] = c + E[\Theta(B)\varepsilon_t] = c$, in the general case, let's

$$\mu = \frac{c}{1 - \phi_1 - \cdots - \phi_p}.$$

It then follows that $\Phi(B)(Y_t - \mu) = \Theta(B)\varepsilon_t$.

Hence $E[Y_t] = \mu$. Note that using stationarity we can retrieve this formula.

4. In the general case, the average is $\mu = \frac{c}{1-\phi}$, and we have proved that $Z_t = Y_t - \mu$ follows the equation $Z_t - \phi Z_{t-1} = w_t$ which has mean 0 and same variance as Y , so that:

$$Z_t^2 = (\phi Z_{t-1} + w_t)^2 = \phi^2 Z_{t-1}^2 + 2\phi w_t Z_{t-1} + w_t^2,$$

leading to:

$$E[Z_t^2] = \phi^2 E[Z_{t-1}^2] + E[w_t^2],$$

as the process is stationary, $\gamma_0 = \sigma^2(1 - \phi^2)^{-1}$.

Using the recurrence equation that defines Y , we can show (see notes) that:

$$\gamma_k = \phi \gamma_{k-1},$$

which is enough to prove the result.

5. The rescaled variance is given by:

$$\begin{aligned} n\text{var}(\bar{Y}_n) &= n \sum_{i,j=1}^n \text{cov}(Y_i, Y_j) \\ &= n \frac{\gamma_0}{n} \left(1 + \frac{2}{n} \sum_{k=1}^{n-1} (n-k) \rho_k \right) \\ &= \gamma_0 + 2 \sum_{k=1}^n (1 - k/n) \gamma_k. \end{aligned}$$

If the series follows an AR(1), then $\gamma_k = \sigma^2 \phi^k / (1 - \phi_1^2) = \gamma_0 \phi^k$, for $k > 0$. Thus,

$$\begin{aligned} n\text{var}(\bar{Y}_n) &= \gamma_0 + 2\gamma_0 \sum_{k=1}^n (1 - k/n) \phi_1^k \\ &= \gamma_0 + 2\gamma_0 \sum_{k=1}^n \phi_1^k - \frac{2\gamma_0}{n} \sum_{k=1}^n k \phi_1^k, \end{aligned}$$

By taking the limit when $n \rightarrow \infty$, we get

$$\gamma_0 + 2\gamma_0 \sum_{k=1}^{\infty} \phi_1^k = \gamma_0 + 2\gamma_0 \left(\sum_{k=0}^{\infty} \phi_1^k - 1 \right) = \gamma_0 \frac{1 + \phi}{1 - \phi}.$$

□